Question 1.

$R_4 \leftrightarrow R_1$

$-R_1 + R_4 \to R_4$

$-2R_1 + R_3 \to R_3$

$$\begin{bmatrix} 1 & 14 & -6 & 2 \\ 3\rho & 5 & -9\gamma & 1 \\ 0 & -28+\beta & 15 & -4+\rho^2 \\ 0 & -14+\rho & 6-\gamma & -2+\rho \end{bmatrix}$$

a) $\rho = 0 \quad \gamma = 0 \quad \beta = 0 \quad k = -14 \quad m = 5$

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Question 2.

a) There are 2 unknowns (x, y) and there will be min. 2 column spaces.

So that dimension of the space will be min. 2

b) $x + 2y + z = 4$ → proposition

c) Yes it forms a vector space. 3 dimensions in 3 dimensional space.

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Question 3.

a) Let $H = \begin{bmatrix} x & a & b \\ a & y & c \\ b & c & z \end{bmatrix}$ $\quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$H^2 = I$ because H is also identity matrix, like $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

b) $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$|v_1| = \sqrt{1^2+0^2+0^2} = 1$ $\quad |v_2| = 1$ $\quad |v_3| = 1$ → so that H is orthonormal matrix.

$v_1 \perp v_2 \perp v_3$

d) Yes, H is diagonalizable. Because eigen values are different.

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Question 4.

a) $$\begin{bmatrix} V_{k+1} \\ W_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} V_k \\ W_k \end{bmatrix}$$

The water has to go W to V. W will be decrease, V willl be Increase.

This equation set describes the situation.

b) tank 1 $v(1) = 25$ liters (10 liquid, 15 water)

%40 liquid, %60 water (from second tank)

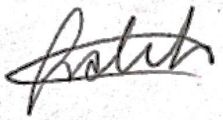
c) tank 1 => 10 liters liquid

tank 2 => 0 liters liquid

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